GATE

Be sure with Vidyalankar Since 1960

Pearl Centre, S.B. Marg, Dadar (W), Mumbai – 400 028. Tel. 4232 4232

CS : COMPUTER SCIENCE AND INFORMATION TECHNOLOGY
Computer Networks

Index

Sr. No.	Contents	Topics	Pg. No.
1. Introduction to ISO/OSI Stack, Physical Layer			
	Notes	Introduction	1
		Network Hardware	4
		Line Configuration	5
		Topology	6
		Comparison between Bus Network and Ring Network	10
		Transmission Mode	11
		Design Issues of Layer	12
		Reference Models	13
		Comparison of OSI and TCP / IP Reference Models	19
		Physical Layer	20
		ISDN	30
		Comparison of ISDN PBX and LAN	32
		Communication Satellite	32
		LMR (Last Minute Revision)	34
	Assignment–1	Questions	37
	Test Paper–1	Questions	40
2. Data Link Layer			
	Notes	Introduction	43
		Framing	45
		Error Detection and Correction	48
		Access Control	50
		Flow Control	52
		Error Control	55
		Data Link Protocols	63
		Local Area Networks	68
		Multiple Access Protocols	71
		IEEE 802.3 and Ethernet	74
		IEEE 802.4 Token Bus	78
		Fibre Distributed Data Interface (FDDI)	80
		LMR (Last Minute Revision)	83

Sr. No.	Contents	Topics	Pg. No.
	Assignment–2	Questions	86
	Test Paper–2	Questions	89
3. Network Layer			
	Notes	Introduction	91
		Circuit Switching Networks	92
		Packet Switching Networks	96
		Message Switching	99
		Comparison	100
		Network Layer Protocols	101
		Congestion Control	104
		Connecting Devices	107
		Routing Algorithm	115
		X.25	119
		LMR (Last Minute Revision)	123
	Assignment–3	Questions	126
	Test Paper–3	Questions	129
4. Transport Layer			
	Notes	Introduction	132
		Duties of the Transport Layer	132
		Connection	135
		OSI Transport Protocol	137
		TCP/IP Protocol Suite	138
		Addressing	140
		IP Addressing	141
		Multihomed Device	144
		Special Address	144
		Subnetting and Supernetting	146
		IP Datagrams and Routing	148
		Routing Protocols	150
		Internet Protocol	152
		IP Encapsulation	154
		Internet Control Message Protocol (ICMP)	156
		UDP	160
		LMR (Last Minute Revision)	162
	Assignment–4	Questions	166
	Test Paper–4	Questions	169

Sr. No.	Contents	Topics	Pg. No.
5. Session, Presentation and Application Layers			
	Notes	Session Layer	171
		Presentation Layer	173
		Application Layer	182
		LMR (Last Minute Revision)	190
	Assignment–5	Questions	192
	Test Paper–5	Questions	194
Practice Problems		Questions	196
SOLUTIONS			
Assignments		Answer Key	201
		Model Solutions	203
Test Papers		Answer Key	218
		Model Solutions	220
Practice Problems		Answer Key	235
		Model Solutions	236

Topic 1 : Introduction to ISO/OSI Stack, Physical Layer

INTRODUCTION

The merging of computers and communications had a profound influence on the way computer systems are organized. The concept of the “computer centre” as a room with a large computer to which users bring their work for processing is now totally obsolete. The old model of a single computer serving all of the organization’s computational need has been replaced by one in which a large number of separate but interconnected computers do the job these systems are called “Computer Network”.

Uses of Computer Network :

1. Business Applications
2. Home applications
3. Mobile users
4. Social issues

1. Business Applications

Many companies have a substantial number of computers. For example, a company may have separate computers to monitor production, keep track of inventories and do the payroll. Initially, each of the computers may have worked in isolation from the others, but at some point, management may have decided to connect them to be able to extract and correlate information about the entire company.

The issue here is resource sharing and the goal is to make all programs, equipment and especially data available to anyone on the network without regard to the physical location of the resource and the user. An obvious and widespread example is having a group of office workers to share a common printer. None of the individuals really needs a private printer, besides a high-volume networked printer is often cheaper, faster and easier to maintain than a large collection of individual printers.

One can imagine a company’s information system as consisting of one or more databases and some number of employees who need to access them remotely. The data are stored on powerful computers called servers. Often these are centrally housed and maintained by a system administrator. In contrast, the employees have simpler machines called clients, with which they access remote data.

2. Home Applications

Some of the more popular uses of the internet for home users are as follows :

1. Access to remote information
2. Person-to-person communication
3. Interactive entertainment
4. Electronic commerce

Access to remove information comes in many forms. It can be surfing the World Wide Web for information. Information available includes the arts, business, cooking, government, health, history, hobbies, recreation, science, sports, travel and many others.

All of the above applications involve interactions between a person and remote databases full of information. The second broad category of network use is person-to-person communication. Another type of person-to-person communication often goes by the name of peer-to-peer communication.

Other communication – oriented applications include using the internet to carry telephone calls, video phone and internet radio.

3. Mobile Users

Mobile computers, such as laptops and personal digital assistants (PDA), are one of the fastest growing segments of the computer industry. Many owners of these computers have desktop machines back at the office and want to be connected to their home base even when away from home. Since having a wired connection is impossible in cars and airplanes, there is lot of interest in wireless networks.

Although wireless networking and mobile computing are often related, they are not identical, as figure below shows.

Wireless	Mobile	Applications
No	No	Desktop computers in offices
No	Yes	A notebook computer used in a hotel room
Yes	No	Networks in older, unwired buildings
Yes	No	Portable office; PDA for store inventory

Fig. : Combination of wireless networks and mobile computing.

4. Social Issues

The widespread introduction of networking has introduced new social, ethical and political problems. A popular feature of many networks is newsgroups or bulletin boards whereby people can exchange messages with like minded individuals.

Computer networks offer the potential for sending anonymous messages. In some situations, this capability is desirable. For example, it provides a way for students, soldiers, employees and citizens to blow the whistle on illegal behaviour on the part of professors, officers, superiors and politicians without fear of reprisals.



A Network is a set of devices (often referred to as nodes) connected by media links.

- A node can be a computer, printer or any other device capable of sending and/or receiving data generated by other nodes on the network.

  The links connecting the devices are called communication channels.
  For example, fibre optic link, satellite link etc.

- The old model of a single computer serving all of the organization's computational needs has been replaced by one in which a large number of separate but interconnected computers do the job. These systems are called **computer networks**.

Computer Networks:



Computer network means an interconnected collection of autonomous computers capable of having interconnections with each other.

- Computer networks are generally organized as a series of layers or levels, each one built upon the one below it.

Need of Layered Structure

The need for a layered structure arises from the following advantages that it provides :

(i) ***Reduction in design complexity*** **:**
The whole architecture can be divided into modules (layers) and each module can be designed separately by a group of experts skilled in that area.

(ii) ***Getting the best expertise*** **:**
In layered structure it is not necessary that a designer must know everything about the computer network. An expert in physical communication can design physical layer whereas a software programmer can handle the application layer part.

(iii) ***Separation of functions*** **:**
Analysis and understanding of the architecture becomes easier.

(iv) ***Each layer provides services to the upper layer*** **:**
Upper layers have nothing to do with the methods used by the lower layers for providing the services. This makes implementation of each layer independent of the other. The services are provided by a lower layer through a set of standard interfaces.

(v) ***Implementation Independency*** **:**
Due to implementation independence, changes can be made in any layer without affecting the other layers. This implies that architecture moves along with the technology.

(vi) ***Work Distribution*** **:**
Work gets distributed among the layers. For example, the user interfaces will be handled by upper layers, error and flow control by mid layer and actual transmission by lower levels. This helps in getting the work done systematically.

(vii) ***Flexibility*** **:**
Between each pair of adjacent layers there is an interface which defines primitive operations and services the lower layer offers to the upper layer. This makes possible to have different protocols in different layer; thereby providing flexibility in choosing the best protocol available for a particular implementation.

(viii) ***Clear–cut Interface*** **:**

Clear–cut interfaces also makes it simple to replace the implementation of one layer with a complete different implementation because all that is required for the new implementation is that it offers exactly the same set of services to its upstairs neighbour as the old implementation.

For example, if the communication medium is to be changed from copper wire to fiber optics, then only the physical layer implementation has to be changed. This does not affect the upper layers as physical layer continues to provide the same services in the same manner

NETWORK HARDWARE

1. Broadcast Networks



Broadcast networks have a single communication channel that is shared by all the machines on the network.

- Short messages, called packets sent by any machine are received by all the others. An address field within packet specifies for whom it is intended.
- Machine checks the address field on receiving. Its processes the packet if intended for itself otherwise just ignores it.
- Broadcast systems allow the possibility of addressing a packet to all destinations by using a special code in the address field. When a packet with this code is transmitted, it is received and processed by every machine on the network.
- Some broadcast system also support transmission to a subset of the machines, known as **multicasting**. We can also sent message to all machines. This type of transmission is called broadcast.
  For example: Bus Network.

2. Point-to-Point Networks

- Point-to-point networks consist of many connections between individual pairs of machines.
- To go from the source to the destination, a packet on this network may have to first visit one or more intermediate machines.
- The connection of two or more networks is called an internetwork. The world wide internet is example of an internetwork.
  For example: Ring Network.

LAN, MAN, WAN and SAN



LAN

Local Area Network (LAN) are privately-owned networks within a single building or campus of up to a few kilometers in size. For example: Networks within IIT campus.

MAN

Metropolitan Area Network (MAN) is basically a bigger version of a LAN and can support both data and voice. It covers a group of nearby corporate offices or a city and might be either private or public. For example: Network connecting all IITs.

WAN

Wide Area Network (WAN) spans a large geographical area, often a country or continent. It contains a collection of machines intended for running user programs.

In most WANs, the subnet consists of two distinct components : transmission lines and switching elements. Transmission lines move bits between machines. They can be made of copper wire, optical fiber or even radio links. Switching elements are specialized computers that connect three or more transmission lines. When data arrive on an incoming line, the switching element must choose an outgoing line on which to forward them. These switching computers have been called by various names in the past; the name router is now most commonly used.

SAN

Another kind of network that we need to be aware of is SANs (System Area Networks). SANs are usually confined to a single room and connect the various components of a large computing system. For example, HiPPI (High Performance Parallel Interface) and Fiber Channel are two common SAN technologies used to connect massively parallel processors to scalable storage servers and data vaults. (Because they often connect computers to storage servers, SANs are sometimes defined as storage area networks.) For example: Organisation connected worldwide.

LINE CONFIGURATION

Line configuration refers to the way two or more communication devices attach to link. There are two possible line configurations:

(i) Point-to-Point

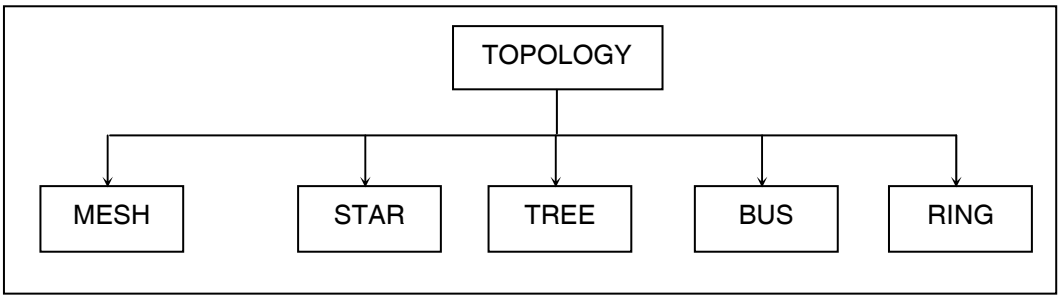
A point-to-point line configuration provides a dedicated link between two devices.

(ii) Multiport

A multiport line configuration is one in which more than two specific devices share a single link.

TOPOLOGY

Topology refers to the way a network is laid out, either physically or logically. Two or more devices connect to a link, then these two or more links form a topology.



- A topology of a network is a geometric representation of the relationship of all the links and linking devices (usually called nodes) to each other.
- Topology is the relative status of the devices to be linked. It can either be peer-to-peer, where the devices share the link equally or primary–secondary, where one device controls traffic and the others must transmit through it.

1. MESH

- In a mesh topology, every device has a dedicated point-to-point link to every other device. The term dedicated means that the link carries traffic only between the two devices it connects. Entire bandwidth is utilized by those two nodes.
- A fully connected mesh network having n devices has n(n–1)/2 physical channels. Thus, every device on the network must have (n–1) input/output (I/O) ports.

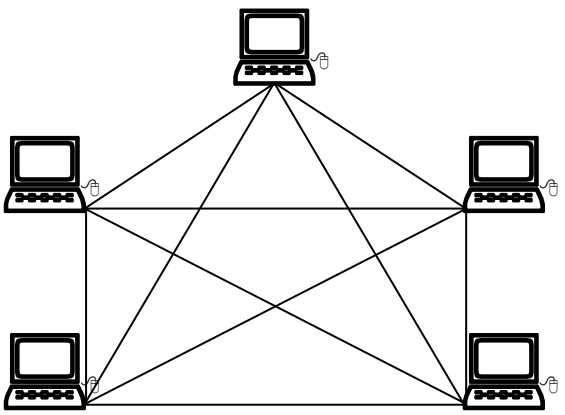


Fig. : Fully Connected Mesh Topology (for five devices)

Advantages

- Guarantees that each connection carry its own data load, thus eliminating the traffic problems that can occur when links are shared by multiple devices.
- If one link becomes unusable, it does not affect the entire system. Thus it is robust.
- It guarantees privacy or security.
- With point-to-point links, fault identification and fault isolation is easy.

Disadvantages

- As every device must be connected to every other device, installation and reconfiguration are difficult.
- The sheer bulk of the wiring can be greater than the available space can accommodate.
- The hardware required to connect each link (I/O port and cable) can be prohibitively expensive.

2. STAR

- In star topology, each device has a dedicated point–to–point link only to a central controller, usually called a hub. Thus the devices are not directly linked to each other.
- If one device wants to send data to another, it sends the data to the controller, and then relays the data to the other connected device(s).

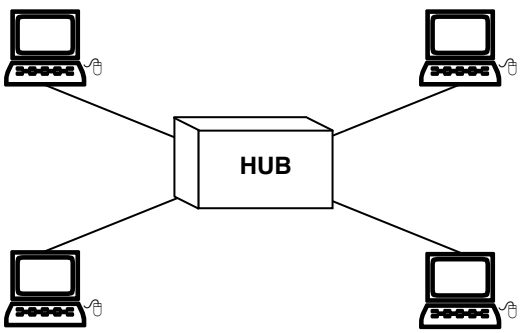


Fig. : Star Topology

Advantages

- Less expensive
- Easy to install and reconfigure
- Less cabling
- Robustness
- Easy fault identification and fault isolation.

Disadvantages

- More cabling is required than some other topologies (such as tree, ring or bus).
- If the hub fails, the entire network goes down.

3. TREE

- Nodes in a tree are linked to a central hub that controls the traffic to the network.
- Not every device plugs directly into the central hub. The majority of devices connect to a secondary hub that in turn is connected to the central hub.

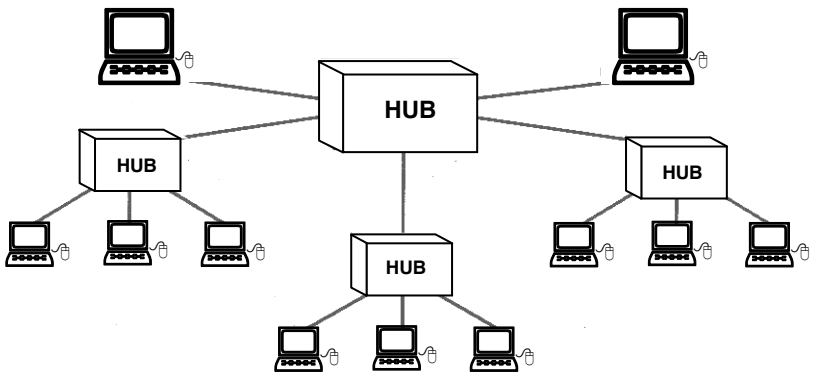


Fig. : Tree Topology

- The central hub in the tree is an active hub, which contains a repeater (a hardware device that regenerates the received bit patterns before sending them out).
- The secondary hubs may be active or passive hubs. Passive hub provides a simple physical connection between the attached devices.

Advantages

- Allows more devices to be attached to a single central hub and therefore increase the distance a signal can travel between devices.
- A allows the network to isolate and prioritize communications from different computers.

Disadvantage

- Not Robust. If central hub goes down, whole system will be affected. More cabling required as compared to bus or ring topology.

4. BUS

- In bus topology one long cable acts as a backbone to link all the devices in the network.

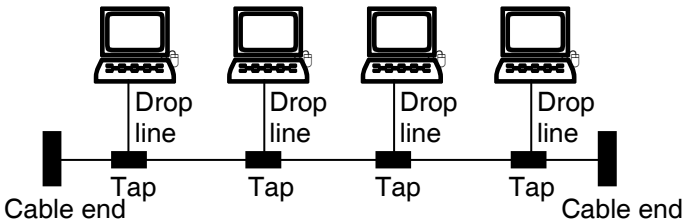


Fig. : Bus Topology

- Nodes are connected to the bus cable by drop lines or taps. A drop line is a connection running between the device and the main cable.
- As a signal travels along the backbone, some of its energy is transformed into heat. (So, it becomes weaker and weaker the farther it has to travel). Thus there is a limit on the number of taps a bus can support and on the distance between those taps.

Advantages

- Ease of installation.
- Uses less cabling than mesh, star or other topologies

Disadvantages

- Difficult reconfiguration and fault isolation.
- Signal reflection at the taps causes degradation in quality.
- A fault or break in the bus cable stops all transmission.

5. RING

- Each device has a dedicated point-to-point line configuration only with the two devices on either side of it. A signal is passed along the ring in one direction, from device to device, until it reaches its destination.
- Each device in the ring incorporates a repeater. When a device receives a signal intended for another device, its repeater regenerates the bits and passes them along.

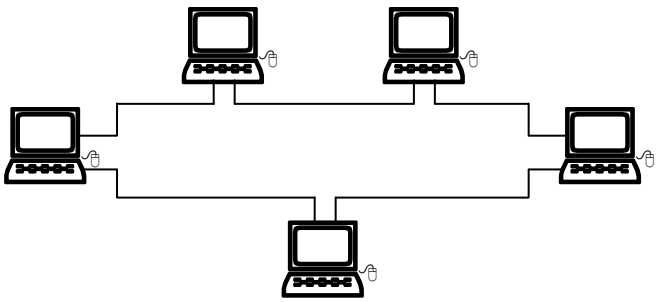


Fig. : Ring topology

Advantages

- Easy to install and reconfigure
- Fault isolation is simplified.

Disadvantages

- A break in the ring can disable the entire network.
- Fault identification is difficult.

Hybrid Topologies

- Combination of several topologies of subnetworks linked together in the larger topology, forms hybrid topology.
- Different topologies are connected to each other via a central controller in a star topology.

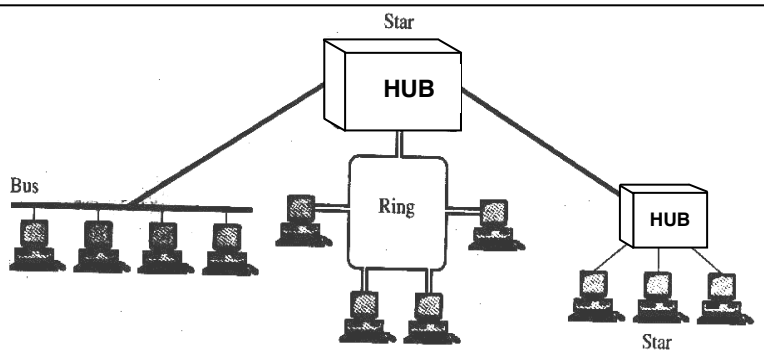


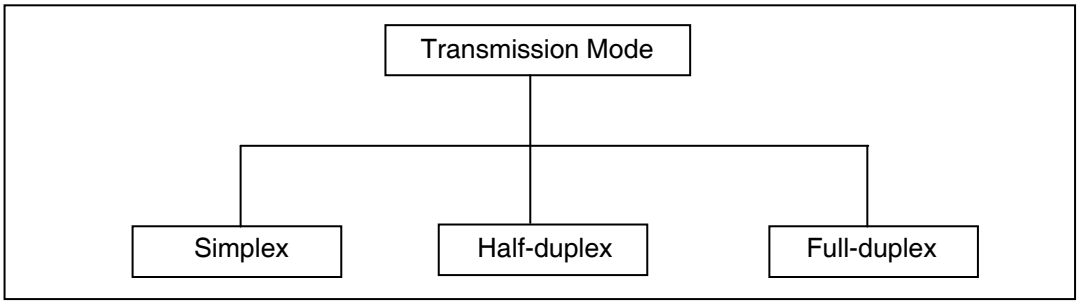
Fig. : Hybrid topology

COMPARISON BETWEEN BUS NETWORK AND RING NETWORK

Comparison Parameters	Bus Network	Ring Network
1. Topology	Broadcast	Point to point
2. Throughput	Determined by the media and access control mechanism	Determined by media and capability of repeater
3. Maximum distance	Low, because high bandwidth is required to support virtual channels	Total span is large, even though intermediable length is less
4. Maximum number of stations	Number of new stations may be added as long as delay and throughput are not affected	It is determined by system design
5. Vulnerability to link or equipment failure	Has a single point of failure, which may take a system vulnerability	It is vulnerable to single break in any node or repeater
6. Message delay	Increases with number of stations and volume of traffic	Increases with number of stations
7. Cost	Cost per node is higher	Is less compared to bus

TRANSMISSION MODE

Transmission mode defines the direction of signal flow between two linked devices. It can either be simplex or half-duplex or full duplex.



Simplex

In simplex mode, the communication is unidirectional. Only one of the two stations on a link can transmit, the other can only receive.

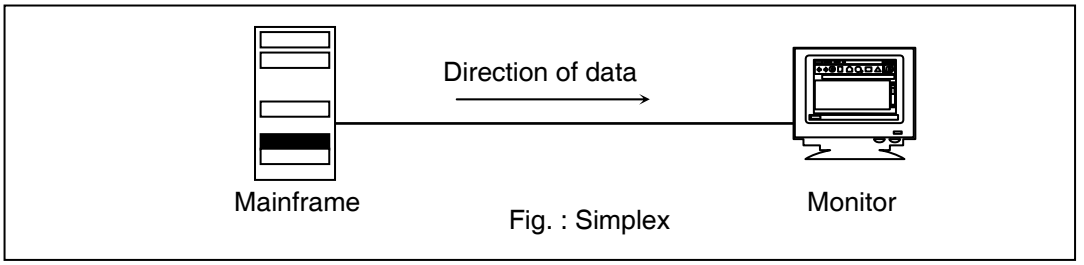


Fig. : Simplex

- Keyboards and traditional monitors are simplex devices. The keyboard can only introduce input, the monitor can only accept output.

Half–Duplex

In half–duplex mode, each station can both transmit and receive, but not at the same time. When one device is sending, the other can only receive, and vice versa.

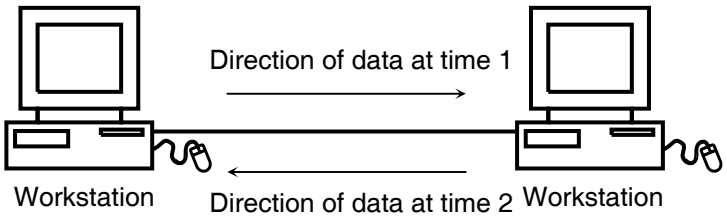


Fig. : Half-duplex

- In a half–duplex transmission, the entire capacity of a channel is taken over by whichever of the two devices is transmitting at the time.
  e.g. : Walky–Talky

Full-Duplex

In full-duplex mode, both stations can transmit and receive simultaneously.

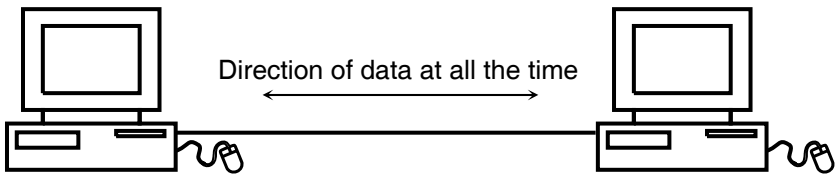


Fig. : Full-duplex

- Signals going in either direction share the capacity of the link. The sharing can occur in two ways, either the link contain two physically separate transmission paths, one for sending and the other for receiving, or the capacity of the channel is divided between signals travelling in opposite directions.
- In a telephone network, when two people are communicating by a telephone line, both can talk and listen at the same time. Thus full-duplex communication.

DESIGN ISSUES OF LAYER

Protocol

A protocol is a set of rules that govern data communication. It also defines what is communicated and when it is communicated.

Internet

A collection of interconnected networks is called an internetwork or just internet.

Intranet

An intranet is a private network that is contained within an enterprise. It may consist of many interlinked Local Area Networks and also use leased lines in the wide area network.

- Every layer needs a mechanism for identifying senders and receivers. A means should be there for a process on one machine to specify with whom it wants to talk.
- Design decisions should concern the rules for data transfer. It can either be simplex or half-duplex or full–duplex communication.
- Error control is an important issue and any error-detecting or correcting codes must be known on both ends of the connection.
- Protocol that make explicit provision for the receiver to allow the pieces to be put back together properly.
- An issue that occurs at every level is how to keep a fast sender from swamping a slow receiver with data.
- Issue that allow an arbitrarily long message to get accepted by the processes.
- When there are multiple paths between source and destination, a route must be chosen. Sometimes this decision split over two or more layers.

Connection-Oriented and Connectionless Services

- In connection–oriented network service, the service user defined establishes a connection, uses the connection and then releases the connection. For example: telephone connection.
- In connectionless service, each message (letter) carries the full destination address and each one is routed through the system independent of all the others. When two messages are sent to the same destination, the first one sent will be the first one to arrive. It is also possible that the first one sent can be delayed so that the second one arrives first.

REFERENCE MODELS

The OSI Reference Model

The International Standards Organization (ISO) is a multinational body dedicated to worldwide agreement on international standards. An ISO standard that covers all aspects of network communication is **Open Systems Interconnection (OSI)** model. The purpose of OSI model is to open communication between different systems without requiring change to the logic of the underlying hardware and software.

Organization of the Layers

The Open System Interconnection model in a layered framework for the design of network systems that allows communication across all types of computer systems. It consists of seven separate but related layers each of which defines a segment of the process of moving information across a network.

The principles that were applied to arrive at the seven layers can be briefly summarized as follows:

1. A layer should be created where a different abstraction is needed.
2. Each layer should perform a well defined function.

3. The function of each layer should be chosen with an eye toward defining internationally standardized protocols.
4. The layer boundaries should be chosen to minimize the information flow across the interfaces.
5. The number of layers should be large enough that distinct functions need not be thrown together in the same layer out of necessity and small enough that the architecture does not become unwidely.

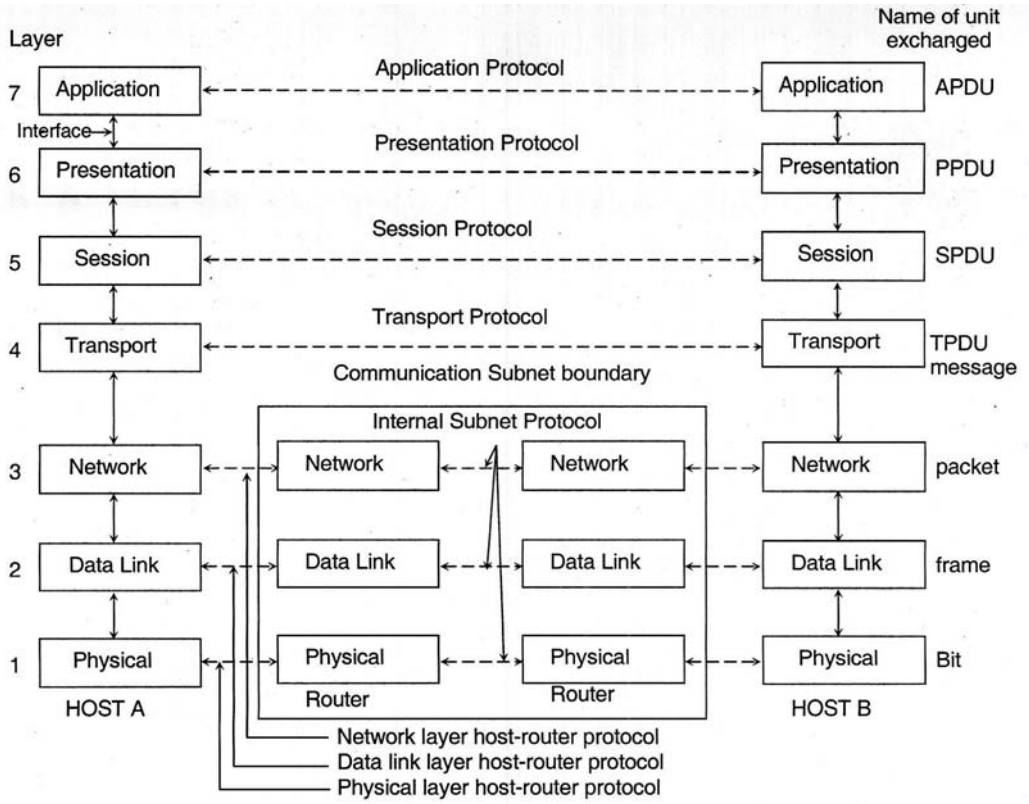


Fig. : The OSI reference model

Functions of the Layer

1. Physical layer :

- The physical layer co-ordinates the functions required to transmit a bit stream over a physical medium. It deals with the mechanical and electrical specifications of the primary connections, such as cables, connections and signaling options that physically link two nodes on a network.
- It oversees the transmission of raw bits over a communication channel.
  For example : Hubs, Repeaters

2. Data link layer

- It accepts a data unit from the third layer and adds meaningful bits to the beginning (header) and end (trailer) that contain address and control information. The data unit with this additional information is called a frame.
- The main task of the data link layer is to transform a raw transmission facility into a line that appears free of undetected transmission errors to the network layer. It accomplishes this task by having the sender break up the input data into data frames (typically a few hundred or a few thousand bytes) and transmit the frames sequentially. If the service is reliable, the receiver confirms the correct receipt of each frame by sending back an **acknowledgement frame**.
- Another issue that arises in the data link layer (and most of the higher layers as well) is how to keep a fast transmitter from drowning a slow receiver in data. Some traffic regulation mechanism is often needed to let the transmitter know how much buffer space the receiver has at the moment.
- It oversees the node-to-node delivery of packets.
- Hence it provides flow control, error control and synchronization.
  For example : Switches, Bridges

3. Network layer :

- The network layer is concerned with controlling the operation of the subnet.
- It oversees source to destination delivery of a packet across multiple network links.
- It provides source and destination address in header.
- It provides functions for routing and multiplexing.
- It is also responsible for congestion control.
  For example : Routers

4. Transport layer :

- The basic function of transport layer is to accept data from session layer, split it up into smaller units if needed and pass these to the network layer.
- It is responsible for source to destination delivery of the entire message.
- It provides service point addressing i.e. delivery of a messages to the appropriate application on a computer running many applications.
- It also provides segmentation and reassembly.

5. Session layer :

- The session layer is the network dialog controller.
- It establishes, maintains and synchronizes the interaction between communicating devices.

  e.g. It might manage an audio stream and video stream that are being combined in a teleconferencing application.

  Specific responsibilities of session layer includes :

  (i) Session management (ii) Synchronization (iii) Dialog control
  (iv) Graceful close of the sessions (v) Token management

6. Presentation layer :

- The presentation layer ensures interoperability among communicating devices.
- It provides the necessary translation of data needed for transmission and at receiver changing that format into the one that is understood by the receiver.
- It provides encryption and decryption for security purpose.

- It provides compression and decompression to make transmission more efficient.

  In particular, presentation layer is concerned with the syntax and semantics of the information transmitted.

7. Application layer :

- The application layer enables the user, whether human or software, to access the network.
- It provides user interfaces and support for services such as electronic mail, remote file access and transfer, shared database and other types of distributed information services.

Layer #	Layer Name	Header	Protocols/Services	Remember This
7	Application (All)	Protocol Data Unit	Telnet, FTP, SMTP, HTTP, File and Print, Email, (WWW), EDI	Where the user requests network services. This is not the application, as in a program. This is database and application services.
6	Presentation (People)	Protocol Data Unit	ASCII, EBCDIC, TIFF, JPEG, GIF, PICT, MIDI, MPEG, QuickTime	Decides how data is represented and translated. Data is formatted for "presentation" to the layers above and below. Encryption, compression and translation take place here.
5	Session (Seem)	Protocol Data Unit	RPC, ZIP, SCP, SQL, X Window, NetBIOS, NFS, ASP, DNA SCP	Establishes, maintains and manages communication sessions between computers. Dialog control occurs here.
4	Transport (To)	Segments	TCP, NBP, UDP, NCP, SPX, ADP, Windowing, flow control, synchronization	Provides reliable data segment transmission. Sets the stage for data disassembly and assembly, before and after transmission. Remember: end-to-end connectivity.

3	Network (Need)	Datagrams or Packets	IP, IPX, BootP, DHCP, ICMP, BGP, OSPF, RIP	If it has anything to do with routing, this is where it happens. This layer determines how data will be routed across a network. Structure and logical (IP) addressing occurs at this layer. Routers operate here.
2	Data LinkSublayers are MAC and LLC (Data)	Frames	MAC, LLC, Frame Relay, LAPB, PPP, calculating CRC or FCS, controls access to the physical medium	This layer is concerned with the links and mechanisms that move data. Topology (Ethernet or Token Ring) is defined here. Switches (generally) and all bridges operate here. Remember: Framing.
1	Physical (Processing)	Bits (1s and 0s)	Ethernet, Token Ring, HSSI, 802.3, bit synchronization, physical connector specifications	If it's on a network and you can touch it, it's here. This layer handles the electrical and physical specifications for network media that carry data bits across a network. Hubs, repeaters and multiplexers operate here.

Overview of OSI layers

TCP/IP Reference Model



The **Transmission Control Protocol / Internetworking Protocol (TCP/IP)** is a set of protocols, or a protocol suite, that defines how all transmissions are exchanged across the internet.

- TCP/IP was designed before OSI model. The layers in the TCP/IP protocol suite do not match exactly with those in the OSI model.
- The TCP/IP protocol suite is made of five layers : Application, transport, internet, host to network. Host to network is divided into (i) Datalink (ii) Physical.
- A network in a TCP/IP internetwork can be a Local Area Network (LAN), a Metropolitan Area Network (MAN), or a Wide Area Network (WAN).

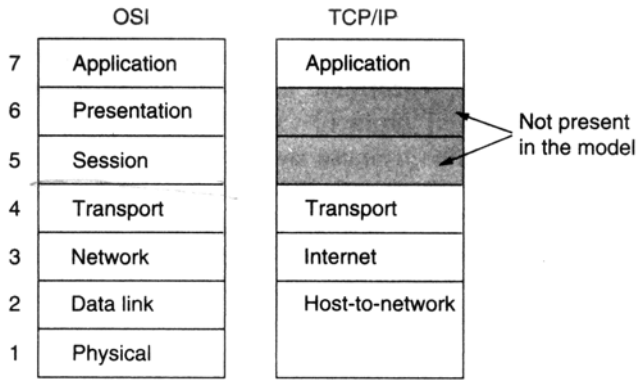


Fig. : The TCP/IP reference model

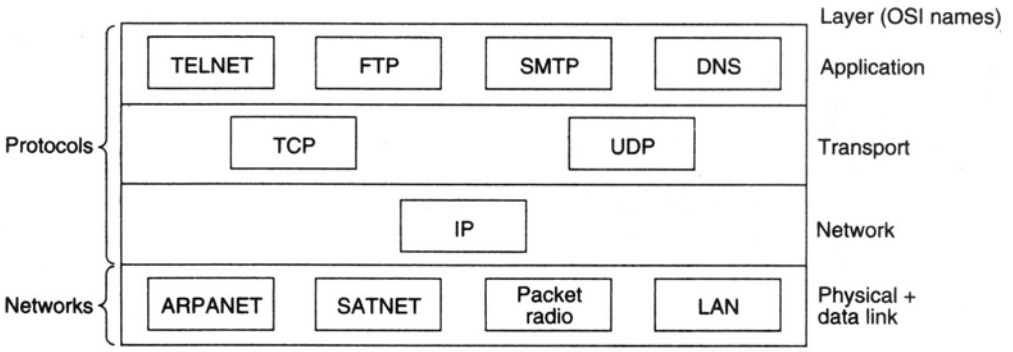


Fig. : Protocols and networks in the TCP/IP model initially

Functions of the Layers

1. Host to Network Layer

- It consists of Physical and Data Link Layers
- At the physical and DLL, TCP/IP does not define any specific protocols. It supports the protocols defined by the underlying networks.
- A network in TCP/IP internetwork can be a Local Area Network (LAN), a Metropolitan Area Network (MAN) or a Wide Area Network (WAN).

2. Internet Layer

- At the internet layer, TCP/IP supports the internet protocol (IP). IP contains four supporting protocols : ARP, RARP, ICMP and IGMP.
- The Internet Protocol (IP) is the transmission mechanism used by the TCP/IP protocols. It is an unreliable and connectionless datagram protocol.
- IP transports data in packets called datagrams, each of which is transported separately.
- Address Resolution Protocol (ARP) is used to translate the protocol address to an equivalent physical hardware address.

- Reverse Address Resolution Protocol (RARP) is used to find IP address of the station or node when its physical address is known.
- Internet Control Message Protocol (ICMP) is a mechanism used by hosts and routes (gate ways) to send modification of datagram problems back to the sender.
- Internet Group Message Protocol (IGMP) is used to help simultaneous transmission of a message to a group of recipients.

3. Transport Layer

- The transport layer is represented by two protocols : TCP and UDP. Both are responsible for delivery of a message from source application to destination application.
- User datagram protocol (UDP) is a process to process protocol that adds only port addresses, checksum error controls. UDP is a connectionless protocol.
- UDP (User Datagram Protocol), is an unreliable, connectionless protocol for applications that do not want TCP's sequencing or flow control and wish to provide their own. It is also widely used for one-shot, client-server type request-reply queries and applications in which prompt delivery is more important than accurate delivery, such as transmitting speech or video.
- The Transmission Control Protocol (TCP) provides full transport layer services to application.
- TCP is a reliable connection oriented protocol that allows a byte stream originating on one machine to be delivered without error on any other machine in the internet. It fragments the incoming byte stream into discrete messages and passes each one on to the internet layer. At the destination, the receiving TCP processes reassembles the received messages into the output stream. TCP also handles flow control to make sure a fast sender cannot swamp a slow receiver with more messages than it can handle.

4. Application Layer

- Application layer is equivalent to combined session, presentation and application layers in the OSI model.
- It contains all the higher level protocols. The early ones included virtual terminal (TELNET), file transfer (FTP), and electronic mail (SMTP). The virtual terminal protocol allows a user on one machine to log onto a distant machine and work there. The file transfer protocol provides a way to move data efficiently from one machine to another. Electronic mail was originally just a kind of file transfer, but later a specialized protocol (SMTP) was developed for it.

COMPARISON OF OSI AND TCP/IP REFERENCE MODELS

Similarities between OSI and TCP/IP :

- Both OSI and TCP/IP reference models are based on the concept of a stack of independent protocols.
- In both models, the layers above the transport layer are application oriented users of the transport service.

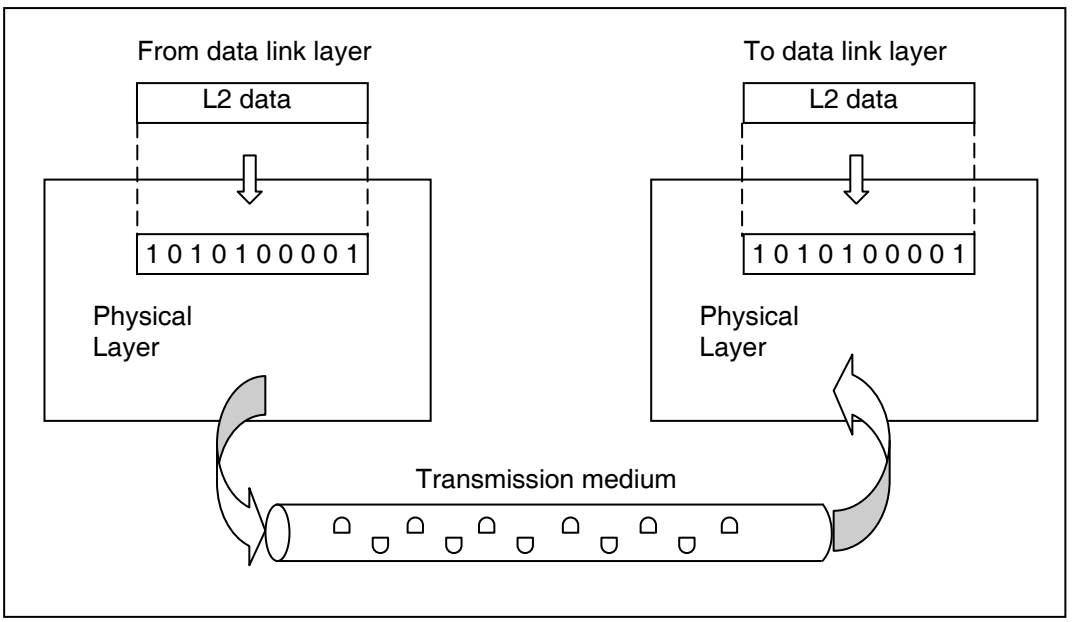
- The service definition tells what the layer does, not how entities above it access it or how the layer works. Each layer performs some services for the layer above it.
- A layer interface tells the processes above it how to access it. It specifies what the parameters are and what results to expect and nothing about how the layer works inside.

Difference between OSI and TCP/IP Reference Model :

- The OSI reference model was devised before the protocol were invented whereas with TCP/IP, the protocols came first and the model was just a description of the existing protocols.
- The OSI model has seven layers and the TCP/IP has five layers. Both have network, transport and application layers, but the other layers are different.
- The OSI model supports connection-oriented communication in the network layer whereas TCP/IP model has only one mode in the network layer i.e. connectionless.
- The OSI model has only connection-oriented communication in the transport layer whereas TCP/IP supports both connectionless and connection-oriented modes in the transport layer, giving the user a choice.

PHYSICAL LAYER

- The physical layer coordinates the function required to transmit a bit stream over a physical medium.
- It deals with the mechanical and electrical specification of the primary connections such as cable, connectors and signaling options that physically link two nodes on a network.



- The first layer receives a data unit from the second layer, put it into a format capable of being carried by a communication link.

Transmission Media

- All telecommunication devices use signals to represent data. These signals are transmitted from one device to another in the form of electromagnetic energy.

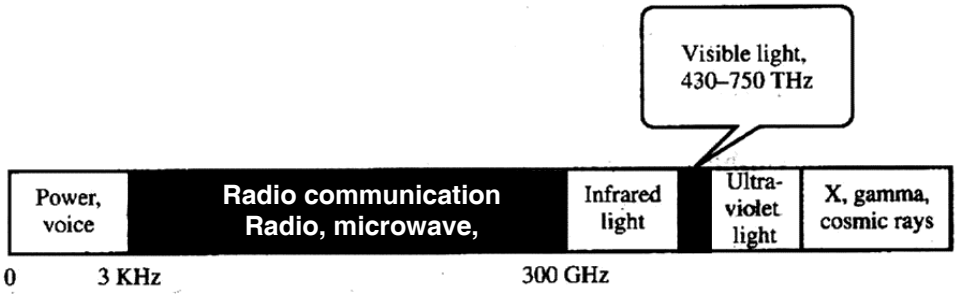
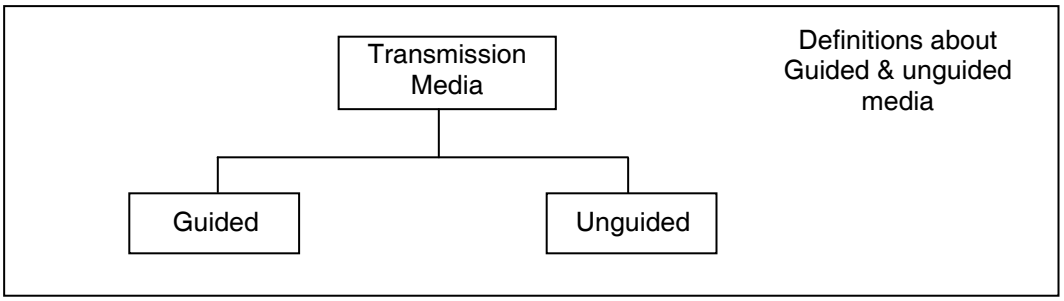
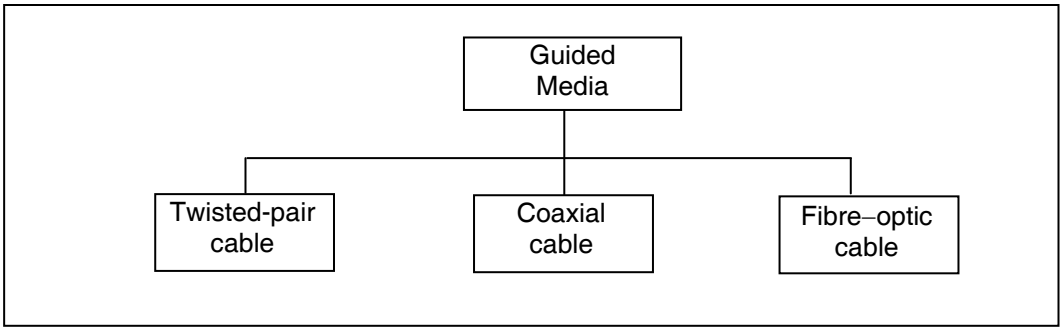


Fig. : Electromagnetic spectrum

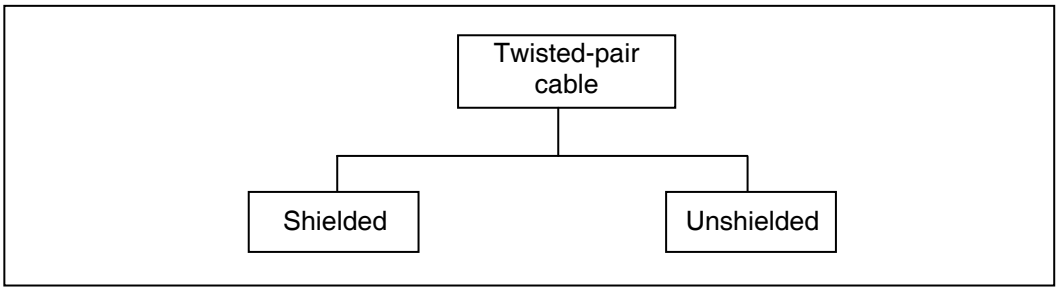
Transmission media can be divided as



Guided Media



Twisted–pair cable



Twisted Pair

The least expensive and most widely used guided transmission medium is twisted pair.

Physical Description

A twisted pair consists of two insulated copper wires arranged in a regular spiral pattern. A wire pair acts as a single communication link. Typically, a number of these pairs are bundled together into a cable by wrapping them in a tough protective sheath. Over longer distances, cables may contain hundreds of pairs. The twisting tends to decrease the crosstalk interference between adjacent pairs in a cable. Neighbouring pairs in a bundle typically have somewhat different twist lengths to reduce the crosstalk interference. On long-distance links, the twist length typically varies from 5 to 15 cm. The wires in a pair have thickness of from 0.4 to 0.9 mm.

Applications

By far the most common transmission medium for both analog and digital signals is twisted pair. It is the most commonly used medium in the telephone network and is the workhorse for communications within buildings.

In the telephone system, individual residential telephone sets are connected to the local telephone exchange or "end office", by twisted-pair wire. These are referred to as subscriber loops.

Transmission Characteristics

Twisted pair may be used to transmit both analog and digital transmission. For analog signals, amplifiers are required about every 5 to 6 km. For digital transmission (using either analog or digital signals), repeaters are required every 2 or 3 km.

Compared to other commonly used guided transmission media (coaxial cable, optical fiber), twisted pair is limited in distance, bandwidth and data rate. As shown in the figure, the attenuation for twisted pair is a very strong function of frequency.

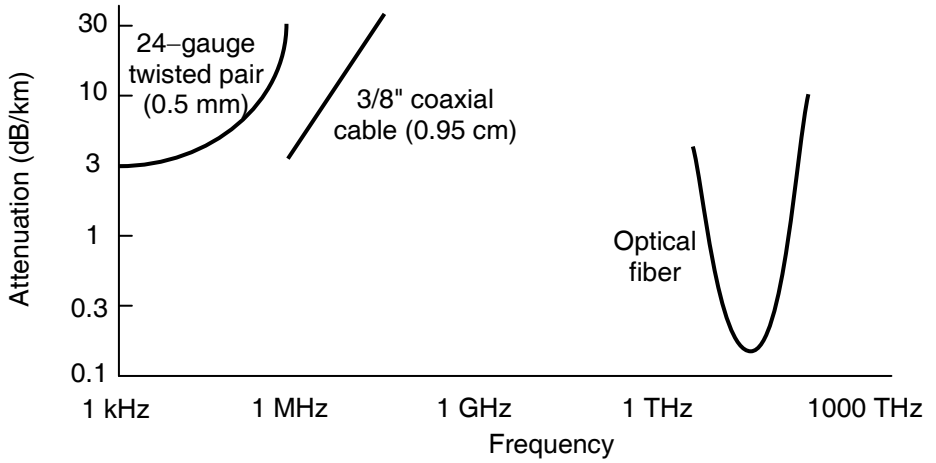


Fig. : Attenuation of Typical Guided Media

Twisted pair comes in two varieties
(1) Unshielded Twisted Pair (UTP) Cable
(2) Shielded Twisted Pair (STP) Cable

Unshielded Twisted-pair (UTP) Cable

- A twisted pair consists of two conductors (usually copper), each with its own colored plastic insulation.
- The plastic insulation is color-banded for identification. Colours are used both to identify the specific conductors in a cable and to indicate which wires belong to pairs and how they relate to other pairs in a larger bundle.

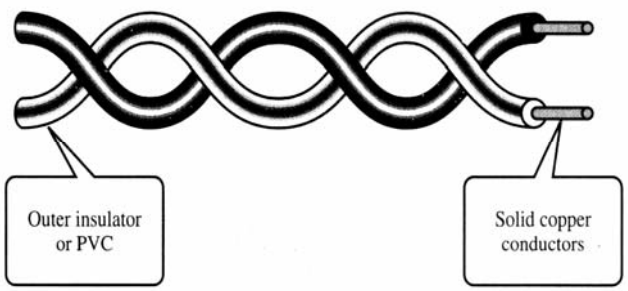


Fig. : Twisted–pair cable

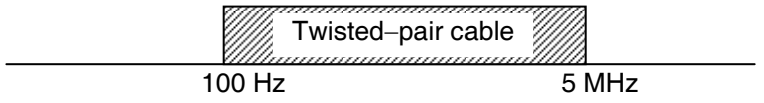


Fig. : Frequency range for twisted-pair cable

Various categories of UTP cable

Category 1 :

→ Used in telephone system

→ Quality is fine for voice but inadequate for all but low–speed data communication.

Category 2 :

→ Suitable for voice and for data transmission of up to 4 Mbps.

Category 3 :

→ Standard cable for most telephone systems.

→ Used for data transmission of up to 10 Mbps

Category 4 :

→ Should have at least three twists per foot as well as other conditions to bring the possible transmission rate to 16 Mbps.

Category 5 :

→ Used for data transmission up to 100 Mbps.
They are similar to category 3 pairs, but with more twists per centimeter, which results in less crosstalk and a better quality signal over longer distances, making them more suitable for high–speed computer communication.

Shielded Twisted-pair (STP) Cable

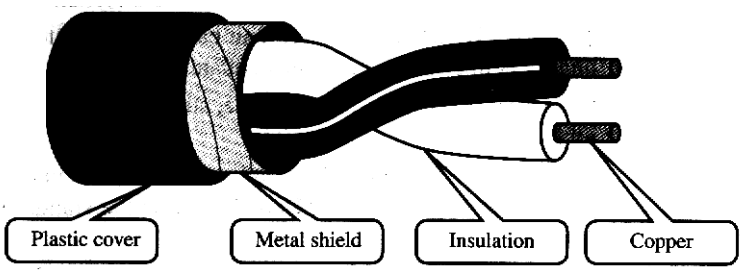


Fig. : Shielded Twisted Pair (STP) cable

- Shielded twisted-pair (STP) cable has a metal foil covering that encases each pair of insulated conductors. This metal casing prevents the penetration of electromagnetic noise.

Advantages of twisted pair cable

1. Low cost
2. Ease of use
3. Easy to Install
4. Used in LAN Technologies.
5. Used in Ethernet and Token ring.

Table : Comparison of Shielded and Unshielded Twisted Pair

Frequency (MHz)	Attenuation (dB per 100 m)			Near–end Crosstalk (dB)		
	Category 3 UTP	Category 5 UTP	150–ohm STP	Category 3 UTP	Category 5 UTP	150–ohm STP
1	2.6	2.0	1.1	41	62	58
4	5.6	4.1	2.2	32	53	58
16	13.1	8.2	4.4	23	44	50.4
25	–	10.4	6.2	–	41	47.5
100	–	22.0	12.3	–	32	38.5
300	–	–	21.4	–	–	31.3



Crosstalk is the undesired effect of one circuit (or channel) on another circuit (or channel).

- Crosstalk occurs when one line picks up some of the signals traveling down another line. Shielding each pair of a twisted-pair cable can eliminate crosstalk.

Coaxial Cable

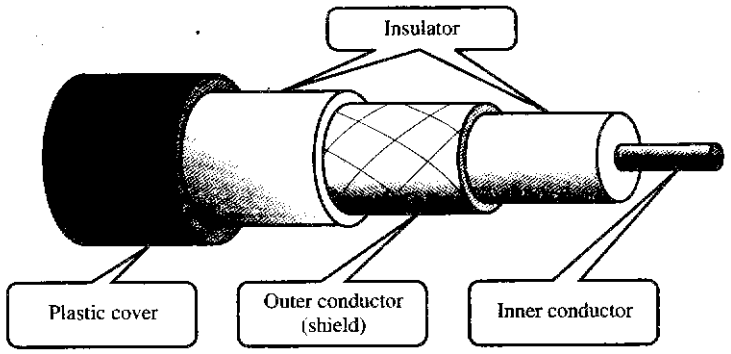


Fig. : Coaxial cable

Physical Description

- Coaxial cable has a central core conductor of solid or wire (usually copper) enclosed in an insulating sheath, which again is encased in an outer conductor of metal foil.
- The outer metallic wrapping works as
  (i) shield against noise
  (ii) second conductor to complete the circuit.
- The outer conductor is enclosed in an insulating sheath and the whole cable is protected by a plastic cover.

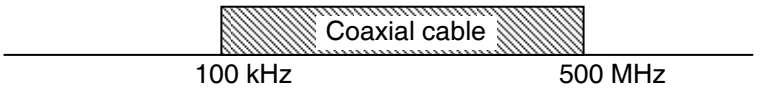


Fig. : Frequency range of coaxial cable

Applications

Coaxial cable is perhaps the most versatile transmission medium and is enjoying widespread use in wide variety of applications. The most important of these are as follows :

- Television distribution
- Long distance telephone transmission
- Short-run computer system links
- Local area networks

Coaxial cable is spreading rapidly as a means of distributing TV signals to individual homes – cable TV.

Coaxial cable is also commonly used for short range connections between devices. Using digital signaling, coaxial cable can be used to provide high-speed I/O channels on computer systems.

Transmission Characteristics

Coaxial cable is used to transmit both analog and digital signals. Coaxial cable has frequency characteristics that are superior to those of twisted pair and can hence be used effectively at higher frequencies and data rates. Because of its shielded, concentric construction, coaxial cable is much less susceptible to interference and crosstalk than twisted pair. The principal constraints on performance are attenuation, thermal noise and intermodulation noise. The latter is present only when several channels (FDM) or frequency bands are in use on the cable.

For long distance transmission of analog signals, amplifiers are needed every few kilometers, with closer spacing required if higher frequencies are used. The usable spectrum for analog signaling extends to about 500 MHz. For digital signaling, repeaters are needed every kilometer or so, with closer spacing needed for higher data rates.

Coaxial Cable Standards

Different coaxial cable designs are categorized by their radio government (RG) ratings. Each RG number denotes a unique set of physical specifications, including the wire gauge of the inner conductor, the thickness and type of the inner insulator, the construction of the shield and the size and type of the out casing. Each cable defined by the RG rating is adapted for a specialized function. RG–8 is used in thick Ethernet. RG – 58 is used in thin Ethernet. RG–59 is used for TV.

Advantages

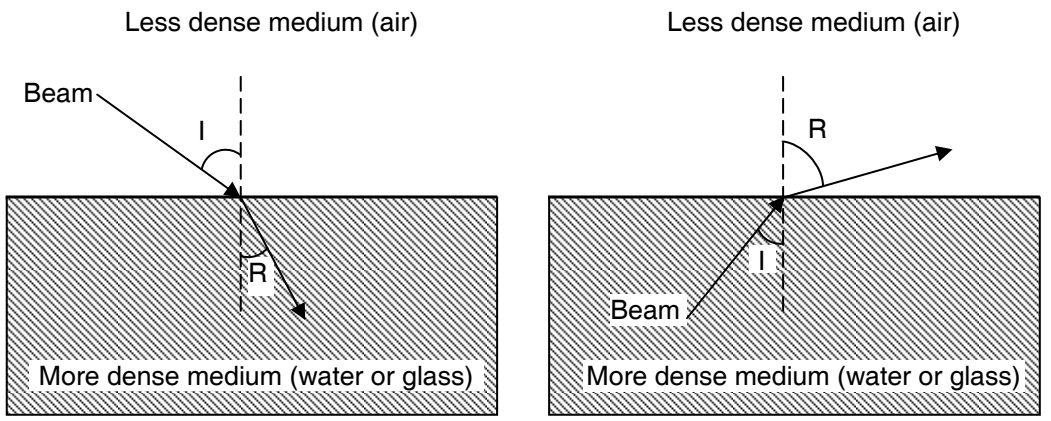
(i) Coaxial cable is used for both data transmission i.e. analog and digital data transmission.
(ii) It has higher bandwidth
(iii) Easy to handle and relatively inexpensive as compared to fiber optic cables.
(iv) It uses for longer distances at higher data rates.
(v) Excellent noise immunity.

Optical Fiber

Optical fiber is made of glass or plastic and transmits signals in the form of light.

- The speed of light depends on the density of the medium through which it is travelling, the higher the density, the slower will be the speed.

If a ray of light traveling through one substance suddenly enters into another (less or more dense) substance, its speed changes abruptly, causing the ray to change direction. This change is called refraction.



(a) From less dense to more dense medium (b) From more dense to less dense medium

Fig. : Refraction

- When light travels into a more dense medium, the angle of incidence is greater than the angle of refraction.
- When light travels into a less dense medium, the angle of incidence is less than angle of refraction.

When a beam of light moves from a more dense into less dense medium and as the angle of incidence increases, so does the angle of refraction. The change in the incident angle results in a refracted angle of 90°. The incident angle at this point is known as the critical angle.

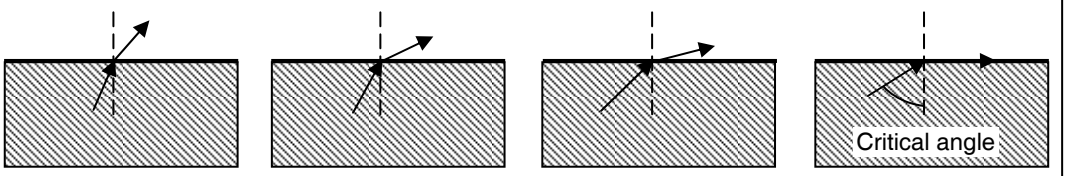


Fig. : Critical angle

- When the angle of incidence becomes greater than the critical angle, the angle of incidence is always equal to the angle of reflection.
  i.e. when the ray of light reflects off a surface, the angle of incidence is equal to the angle of reflection.

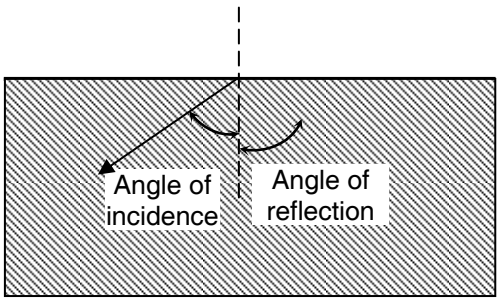
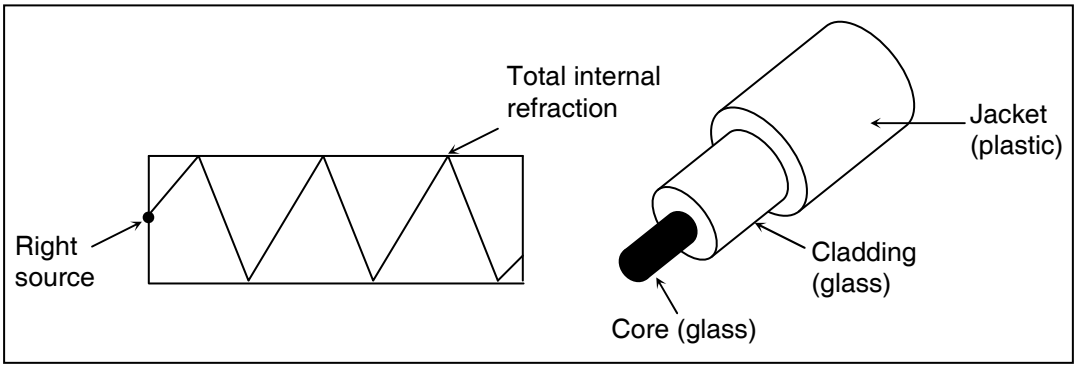


Fig. : Reflection

Optical fiber is made of glass or plastic and transmits signals in the form of light.



Principle on which fiber cables work

When a light ray passes from one medium to another, for example, from fused silica to air, the ray is refracted at the silica/air boundary. For angles of incidence above a certain critical value, the light is refracted back into the silica, none of it escapes into the air. Thus a light ray incident at or above the critical angle is trapped inside the fiber and can propagate for many kilometers with virtually no loss.

Construction

At the center is the glass core through with the light propagates. In multimode fibers, the core is 50 microns in diameter and in case of single mode fibers it is 8 to 10 microns. The core is surrounded by a glass cladding with a lower index of refraction than the core, to keep all the light in the core.

Next comes a thin plastic jacket to protect the cladding. Optical fibers main applications are long haul trunks, Metropolitan trunks, etc.

Total Data Rate	2 Gbps
Bandwidth	2 GHz
Repeater Spacing	10 to 100 km

Maximum Data Rate of the Channel

Nyquist's Theorem

Henry Nyquist proposed a theorem that has had profound effects on information theory as well as the practical design of data communication techniques involving digitalization of analog signals. Also original theorem was regarding analog signals and noiseless channels and later it was applied to digitl signals.

Nyquist proved that if an arbitrary signal has been run through a low pass filter of bandwidth H, the filtered signal can be completely reconstructed by making only 2H samples per second. If the signal consists of V discrete levels,

Nyquist's Theorem states :

Maximum data rate = $2H\log_2 V$ bits / sec

For example : A noiseless 3KHz channel cannot transmit binary signal at a rate exceeding 6000 bps.

Shannon's Theorem

Shannon's major result is that the maximum data rate of a noisy channel whose bandwidth is H Hz, and whose signal-to-noise ratio is S/N, is given by,

Maximum number of bits/sec = $H\log_2 (1+S/N)$

Shannon's result was derived using information theory arguments and applied to any channel subject to Gaussian (thermal) noise.

Table : Point–point Transmission Characteristics of Guided Media

	Frequency Range	Typical Attenuation	Typical Delay	Repeater Spacing
Twisted pair (with loading)	0 to 3.5 KHz	0.2 dB/km @1 KHz	50 μs/Km	2 Km
Twisted pairs (multi–pair cables)	0 to 1 MHz	3 dB/km @1 KHz	5 μs/Km	2 Km
Coaxial cable	0 to 500 MHz	7 dB/Km @ 10 MHz	4 μs/Km	1 to 9 Km
Optical fiber	180 to 370 THz	0.2 to 0.5 dB/Km	5 μs/Km	40 Km

ISDN

Evolution of ISDN

For more than a century, the primary international telecommunication infrastructure has been the public circuit-switched telephone system. This system was designed for analog voice transmission and was inadequate for modern communication needs. Anticipating considerable user demand for an end-to-end digital services, the world's telephone companies and PTTs got together in 1984 under the auspices of CCITT and agreed to build a new, fully digital circuit–switched telephone system by the early part of the 21st century. This new system, called ISDN (Integrated Services Digital Network), has as its primary goal the integration of voice and non-voice services.

ISDN System Architecture

Digital pipe

The key idea behind ISDN is that of the digital bit pipe, a conceptual pipe between the customer and carrier through which bits flow. The bits can flow in both directions. The digital pipe can support multiple independent channels by time division multiplexing of the bit stream. The exact format of the bit stream and its multiplexing is a carefully defined part of the interface specifications for the digital bit pipe.

The figure given below gives the conceptual view of the ISDN connection.

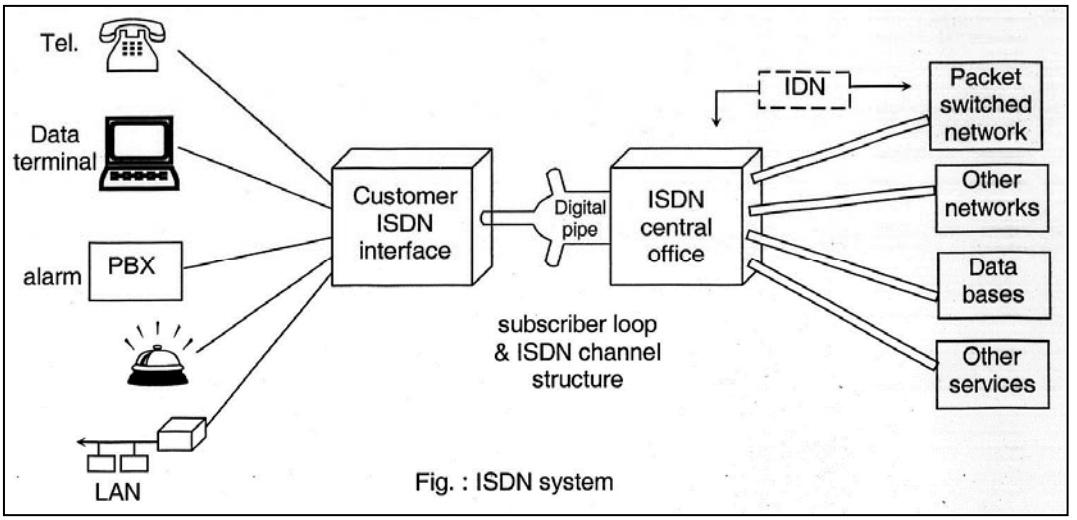


Fig. : ISDN system

1. The carrier places a customer ISDN interface on the customer's premises and connects it to the ISDN exchange in the carriers office, several kilometers away.
2. The subscriber loop under ISDN consists of one or two twisted pair to provide a basic full duplex digital communication link.
3. The customer ISDN interface has a connection into which a passive two cable can be inserted.
   Various devices such as telephones, terminals, alarms can be connected to the cable, similar to the way devices are connected to a LAN. From the customer's point of view, the network boundary is the connector on customer services interface.

CCITT Definition of ISDN

An ISDN is a network, in general evolving from a telephone IDN, that provides end-to-end digital connectivity to support a wide range of services including voice and non-voice services to which users have access by a limited set of standard multi-purpose user network interface.

Applications

1. The key ISDN service will continue to be voice, although many enhanced features are added.
2. Advanced voice service include call forwarding and conference calls worldwide.
3. Advanced non-voice services are remote electricity meter reading, and on–line medical, burglar, and smoke alarms that automatically calls the hospital, police or fire department respectively, and give them addresses to speed up response.
4. Other services provided are :
   Telephone, Data transmission, FAX, music, high speed computer communications, TV conferencing, videophone, TV, email, LANs etc.

Interface

The digital pipe between the central office and the ISDN user will be used to carry a number of communication channels. The capacity of the pipe and therefore the number of channels carried may vary from user to user.

Several channel types have been standardized :

A – 4 kHz analog telephone channel
B – 64 kbps digital PCM channel for voice or data
C – 8 or 16 kbps digital channel
D – 16 kbps or 64 kbps digital channel for out–of–band signaling
E – 64 kbps digital channel for internal ISDN signaling
H – Hybrid channel 384 – 1536 or 1920 kbps digital channel.

Three combinations have been standardized so for :

1. Basic rate : 2B + 1D
2. Primary rate : 23B + 1D (US and Japan)
   or 30B + 1D (Europe)
3. Hybrid : 1A + 1C

Basic Rate channel

The basic rate should be viewed as a replacement for POTS (Plain Old Telephone Service) for home or small business use.

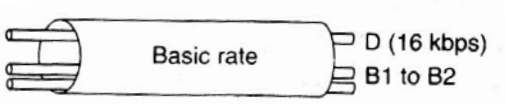


Fig. : Basic rate digital tape

Each of the 64 kbps B channels can handle a single PCM voice channel with 8 bit samples made 8000 times a second. Signaling is on a separate 16 kbps D channel, so the full 64 kbps are available to the user.

Primary rate channel

The primary rate interface was intended for use by large business with a PBX. It has 23B channels and 1D channel in US and Japan and 30 B channels and 1D channel in Europe.

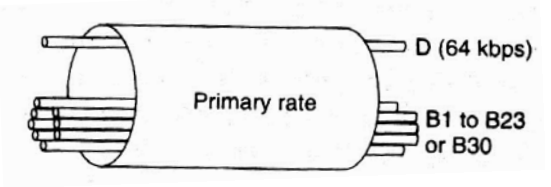


Fig. : Primary rate digital tape

The 23B + 1D choice was made to allow an ISDN frame fit nicely on AT & T's TI system. The 30B + 1 D choice was made to allow an ISDN frame fit nicely in CCITT's 2.048 Mbps system.

COMPARISON OF ISDN PBX AND LAN

ISDN PBX	LAN
1. ISDN terminal with functional groupings of equipments will lead to an ISDN PBX.	LAN is a communication network that provides interconnection of variety of data communicating devices within a small area.
2. The architecture of an subscriber's premises is broken up functionally into groupings separated by reference model.	The architecture of LAN is described in terms of layering of protocols that organize the basic function of LAN
3. CCITT defined the reference model	IEEE 802 reference model is accepted as LAN protocols.
4. Four reference point R, S, T and U are defined. S reference point is the ISDN PBX and the ISDN terminal.	Seven different layers with distinct functions are defined.

COMMUNICATION SATELLITE



In satellite based computer networks, communication between the nodes is accomplished using "radio frequencies".

A satellite based networks has a star topology, satellite being the central hub. If the hub fails the entire communication network comes to a standstill. In view of this, reliability and redundancy considerations are important when designing satellites.

Working of Satellite based Computer Networks

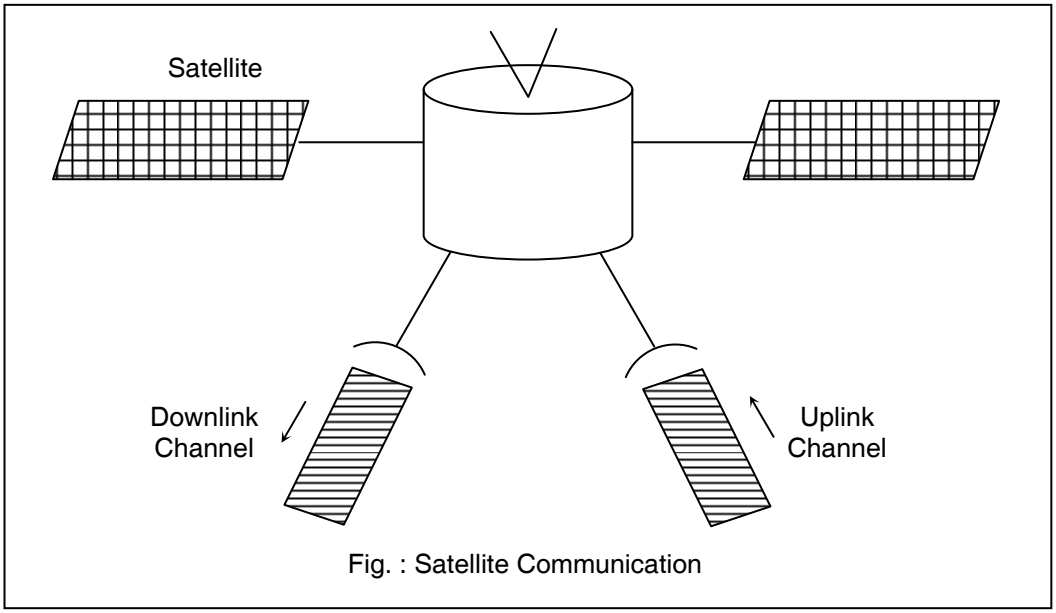


Fig. : Satellite Communication

Communication satellites generally have upto a dozen or so transponders.

- Each transponder has a beam that covers some portion of the earth below it, ranging from a wide beam 10,000 km across to a spot beam only 250 km across.
- Stations within the beam area can send frames to the satellite on the uplink frequency.
  The satellite then rebroadcasts them on the downlink frequency. Different frequencies are used for uplink and downlink to keep the transponder from going into oscillation.



Satellites that do no on-board processing, but just echo whatever they hear are often called bend-pipe satellite.

- Each antenna can aim itself at some area, transmit some frames, and then aim to a new area. Aiming is done electronically, but still takes some number of microseconds.

Protocols : (Channels allocation algorithms)

Just as with LAN's one of the key design issue is how to allocate the transponder channels. However, unlike LANs, carrier sensing is impossible due to the 270 msec propagation delay. Hence the need for other protocols. Five classes of protocols are used on the multiple access (uplink) channel: polling, ALOHA, TDM, CDMA and FDM.

Polling

The traditional way to allocate a single channel among competing users is for somebody to poll them. Having the satellite poll each station in turn to see if it has a frame is prohibitively expensive, given the 270 msec time required for each poll/response sequence. However, if all the ground stations are also tied together to a (typically low bandwidth) packet-switching network, a minor variation of this idea is conceivable.

The idea is to arrange all the stations in a logical ring, so each station knows its successor. Around this terrestrial ring circulates a token. The satellite never sees the token. A station is allowed to transmit on the uplink only when it has captured the token. If the number of stations is small and constant, the token transmission time is short and the burst sent to the uplink channel are much longer than the token rotation time, the scheme is moderately efficient.

Access Algorithms

ALOHA

Two versions of ALOHA are :

(i) Pure ALOHA
(ii) Slotted ALOHA

Using slotted ALOHA doubles the efficiency but introduces the problem of how to synchronize all the stations so they know when each time slot beings. Fortunately, the satellite itself holds the answer, since it is inherently a broadcast medium. One ground station, the reference station, periodically transmits a special signal whose rebroadcast is used by all the stations as the time origin. If the time slots all have length ΔT, each station now knows that time slot k beings at a time $k\Delta T$ after the time origin. Since clocks runs at slightly different rates, periodic resynchronization is necessary to keep everyone in one.

LMR (LAST MINUTE REVISION)

- A Network is a set of devices connected by media links.
- The links connecting the devices are called communication channels.
- Computer network means an interconnected collection of autonomous computers capable of having interconnections with each other.
- Broadcast networks have a single communication channel that is shared by all the machines on the network.
- Point-to-point networks consist of many connections between individual pairs of machines.
- LAN are privately-owned networks within a single building or campus of up to a few kilometers in size.
- MAN is basically a bigger version of LAN and can support both data and voice.
- WAN supports a large geographical area, often a country or continent.
- A line configuration defines the relationship of communication devices to a communication pathway.

- In a multiport line configuration, three or more devices share a link.
- Topology refers to the physical or logical arrangement of a network. Devices may be arranged in a mesh, star, tree, bus, ring or hybrid topology.
- Communication between two devices can occur in one of three transmission modes: simplex, half-duplex or full-duplex.
- In simplex mode, the communication is unidirectional. Only one of the two stations on a link can transmit, the other can only receive.
- In half-duplex mode, each station can both transmit and receive, but not at the same time. When one device is sending, the other can only receive, and vice versa.
- In full-duplex mode, both stations can transmit and receive simultaneously.
- An internet is a network of networks.
- The Transmission Control Protocol/ Internetworking Protocol (TCP/IP) is a set of protocols, or a protocol suite, that defines how all transmission are exchanged across the internet.
- The International Standards Organization (ISO) created a model called the Open Systems Interconnection (OSI), which allows diverse systems to communicate.
- The seven–layer OSI model provides guidelines for the development of universally compatible architecture, hardware and software.
- The physical, data link and network layers are the network support layers.
- The session, presentation and application layers are the user support layers.
- The transport layer links the network support layers and the user support layers.
- The physical layer coordinates the functions required to transmit a bit stream over a physical medium.
- The data link layer is responsible for delivering data units from one station to the next without errors.
- The network layer is responsible for the source-to-destination delivery of a packet across multiple network links.
- The session layer establishes, maintains and synchronizes the interaction between communicating devices.
- The presentation layer ensures interoperability between communicating devices through transformation of data into a mutually agreed upon format.
- The application layer enables the users to access the network.
- Crosstalk is the undesired effect of one circuit (or channel) on another circuit (or channel).
- Optical fiber is made up of glass or plastic and transmits signals in the form of light.
- Information must be transformed into electromagnetic signals prior to transmission across a network.
- Topology refers to the way a network is laid out, either physically or logically. Two or more devices connect to a link, then these two or more links form a topology.

- Transmission mode defines the direction of signal flow between two linked devices. It can either be simplex or half-duplex or full duplex.
- A protocol is a set of rules that govern data communication. It also defines what is communicated and when it is communicated.
- A collection of interconnected networks is called an internetwork or just internet.
- An intranet is a private network that is contained within an enterprise. It may consist of many interlinked local area networks and also use leased lines in the wide area network.
- The service definition tells what the layer does, not how entities above it accessed it or how the layer works. Each layer performs some services for the layer above it.
- A layer interface tells the processes above it how to access it. It specifies what the parameters are and what results to expect and nothing about how the layer works inside.
- Crosstalk is the undesired effect of one circuit (or channel) on another circuit (or channel).
- Optical fiber is made of glass or plastic and transmits signals in the form of light.
- If a ray of light traveling through one substance suddenly enters into another (less or more dense) substance, its speed changes abruptly, causing the ray to change direction. This change is called refraction.
- When a beam of light moves from a more dense into less dense medium and as the angle of incidence increases, so does the angle of refraction. The change in the incident angle results in a refracted angle of 90°. The incident angle at this point is known as the critical angle.
- In satellite based computer networks, communication between the nodes is accomplished using “radio frequencies”.
- Satellites that do no on-board processing, but just echo whatever they hear are often called bend–pipe satellite.
- A satellite based network has a star topology, satellite being the central hub.

❑ ❑ ❑ ❑ ❑ ❑

ASSIGNMENT – 1

Duration : 45 Min. **Max. Marks : 30**

Q1 to Q6 carry one mark each

1. Which out of the following follow primary–secondary relationship between devices?
   (A) Ring (B) Mesh
   (C) Star (D) None of these

2. The Rai Foundation has a fully connected ring network consisting of eight devices, then how many cable links are required.
   (A) 7 (B) 8
   (C) 28 (D) 36

3. Frequency range of coaxial cable is
   (A) 100Hz – 5MHz (B) 100kHz – 500 MHz
   (C) 100 Hz – 500 MHz (D) 100 kHz – 5 MHz

4. Which of the following is the greatest advantage of coax cabling ?
   (A) High security (B) Physical dimensions
   (C) Long distances (D) Easily tapped

5. NNTP
   (A) protocol used for moving news articles around
   (B) protocol used for fetching pages on world wide web
   (C) protocol for mapping host names onto their network addresses
   (D) Allows user on one machine to log onto a distant machine.

6. Suppose BSNL telephone line has a bandwidth of 6100 Hz. The signal–to–noise ratio is 40 dB. Then the capacity of channel would be
   (A) 80,176 bps (B) 38,821 bps
   (C) 32,681 bps (D) 81,056 bps

Q7 to Q18 carry two marks each

7. In a network with 50 computer, a mesh topology would require _____ cables
   (A) 1225 cables (B) 1280 cables
   (C) 50 cables (D) 99 cables

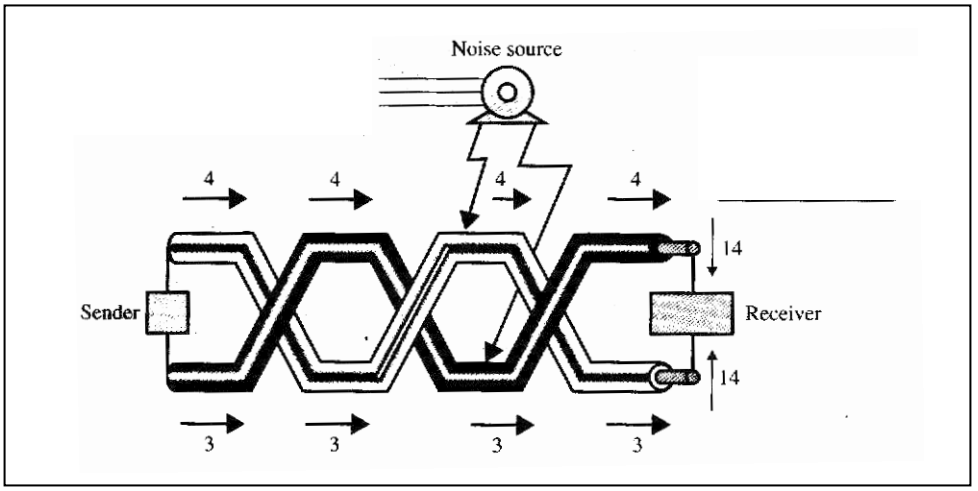
8. Which of the following is true ?
   (A) The OSI model support connectionless communication in the network layer
   (B) The OSI support both connection-oriented and connectionless communication in the transport layer.
   (C) TCP/IP model support connectionless mode in the network layer.
   (D) TCP/IP support connection-oriented mode in transport layer.

9. When a signal travels through a transmission medium, it power becomes 100 times. Then there would be power

(A) Gain of 100dB (B) Loss of 100dB
(C) Gain of 20dB (D) Loss of 20dB

10. Consider two wires are twisted around each other at regular interval. Higher voltage level exist on the wire nearer to noise source as compared to far away wire. The total effect of noise source on the receiver will be

(A) 28 units (B) 1 units
(C) 0 unit (D) 14 units



11. Consider a signal is measured at two different points. The power is P_1 at the first point and P_2 at the second point. If dB is 0, then

(A) P_2 is zero (B) P_2 equals P_1
(C) P_2 is much larger than P_1 (D) P_2 is much smaller than P_1

12. If the angle of refraction is 90 degrees and the angle of incidence is 68 degrees, then the critical angle would be

(A) 22 (B) 68
(C) 90 (D) 158

13. For a particular channel, the bandwidth is 7200 Hz. Signal strength is of 12 μW and one sided power spectrum density of white noise is 10^{-12} Watt/Hz. The channel capacity would be

(A) 29.83 Kbps (B) 77 Kbps
(C) 32.22 Kbps (D) 169 Kbps

14. What is the correct and complete OSI sequence in order from user interface (Layer 7) to the delivery of binary bits (Layer 1)?

(A) Physical Layer, Network Layer, Data Link Layer, Transport Layer, Session Layer, Presentation Layer, Application Layer

(B) Application Layer, Presentation Layer, Session Layer, Transport Layer, Network Layer, Data Link Layer, Physical Layer

(C) Application Layer, Physical Layer, Session Layer, Transport Layer, Network Layer, Data Link Layer, Presentation Layer

(D) Physical Layer, Data Link Layer, Network Layer, Session Layer, Transport Layer, Presentation Layer, Application Layer

15. Given that the channel capacity is 250 Kbps, find the white noise, present in the channel, if the signal strength is, 15μW and bandwidth = 8000Hz, SBR = 20000

(A) 9.375×10^{-11} watts/Hz (B) 9.370×10^{-11} watts/Hz

(C) 9.388×10^{-11} watts/Hz (D) $9.368 \times 10^{-}$watts/Hz

16. Match the columns :

(a) DSI	(i) 51.840Mbps
(b) STS – 24	(ii) 2.488320Gbps
(c) STS – 48	(iii) 1.544Mbps
(d) STS 1	(iv) 1.244160Gbps

(A) a – iv, b – i, c – iii, d – ii (B) a – iii, b – iv, c – I, d – ii

(C) a – iii, b – iv, c – ii, d – I (D) a – iv, b – ii, c, iii, d – i

17. A voice-grade telephone line carries a continuous electromagnetic signals ranging between 300Hz & 3300Hz, what will be the wavelength of 15000Hz, travelling through a copper wire.

(A) 133.33m (B) 13.33m

(C) 13333.33m (D) 333.33m

18. A voice-grade telephone line carries a continuous electromagnetic signals ranging between 300Hz and 3300Hz. What is the channel capacity if the signal to noise ratio is 30dB ?

(A) 29.02kbps (B) 29.2kbps

(C) 28.6kbps (D) 30.13kbps

❑ ❑ ❑ ❑ ❑ ❑

TEST PAPER – 1

Duration : 30 Min. **Max. Marks : 25**

Q1 to Q5 carry one mark each

1. Which of the following is true w.r.t. ISDN channels ?
   (A) A is 64 kpbs digital PCM channel for voice or data
   (B) C is 8 or 16 kbps digital channel
   (C) B is 4 khz analog telephone channel
   (D) H is 64 kpbs digital channel for internal ISDN signaling

2. X.21 is a _____
   (A) Physical layer protocol
   (B) Network layer protocol
   (C) Transport layer protocol
   (D) Session layer protocol

3. The Singh Corporation of Amritsar has a fully connected mesh network consisting of ninety nine devices. Calculate the number of ports for each device.
   (A) 4950 (B) 4851
   (C) 100 (D) 98

4. Voice-band frequencies ranges from
   (A) 0 – 3 kHz (B) 430 THz – 750 THz
   (C) 3 kHz – 300 GHz (D) None of these

5. Frequency range of twisted-pair cable is
   (A) 100 Hz – 300 KHz (B) 0 – 5 MHz
   (C) 100 Hz – 5MHz (D) 300 KHz – 6 MHz

Q6 to Q13 carry one mark each

6. The Shimla Corporation of India has a fully connected mesh network consisting of nine devices. Calculate the total number of cable links needed.
   (A) 36 (B) 45
   (C) 9 (D) 8

7. Suppose MTNL telephone line has a bandwidth of 3000 Hz (300 Hz to 3300Hz). The signal to noise ratio is 3100. Then the capacity of channel would be
   (A) 35,000 bps (B) 34,796 bps
   (C) 34,860 bps (D) 34,660 bps

8. The wavelength of red light whose frequency is 4×10^{14} Hz in air is
   (A) 0.75 μm (B) 7500 pm
   (C) 400 m (D) Data insufficient

9. For a certain medium if critical angle is 60° and the angle of incidence is 70°, the angle of reflection is

(A) 10° (B) 60°
(C) 70° (D) 130°

10. Consider two parallel flat wires one used for communication. High voltage level exist on the wire nearer to noise source as compared to far away wire. The total effect of noise source on the receiver will be

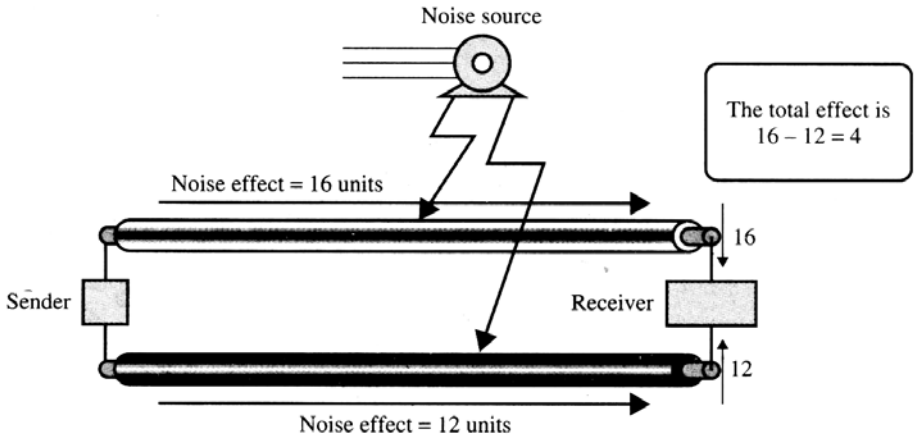


Figure : Communication using a flat wire

(A) 28 units (B) 4 units
(C) 3 units (D) 14 units

11. When a signal travels through a transmission medium, its power becomes one-fifth.
Then there would be power

(A) Loss of 7 dB (B) Gain of 7 dB
(C) Loss of 5 dB (D) Gain of 5 dB

12. For a coaxial optic cable if the propagation speed is 2×10^8 m/s for a distance of 2000 m, then propagation time would be

(A) 10 μ sec (B) 100 μ sec
(C) 0.1 μ sec (D) 0.001 μ sec

13. For a particular channel, the bandwidth is 3600 Hz. Signal strength is 12 μW and two sided power spectrum density of while noise is 2×10^{-12} watt/Hz. The channel capacity would be

(A) 81 Kbps (B) 40 Kbps
(C) 39 Kbps (D) 42 Kbps

Q.14(a) and (b) carry two marks each

Linked Answer Questions

14(a). A certain telecom company 'A' has a line bandwidth of 8000Hz. The signal to noise ratio is 35dB. The capacity of the channel would be

(A) 42860.42bps (B) 28001.099bps
(C) 93017.64bps (D) 80000bps

14(b). Another telecom company 'B' has channel capacity 30% more than that of company A. What will be its SNR?

(A) 48.66 (B) 45.50
(C) 23.81 (D) 38.35

❑ ❑ ❑ ❑ ❑ ❑